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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (Biotechnology) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

20BT505 - BIOETHICS, BIOSAFETY AND IPR

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **Unit – I** and **Two full** questions from **Unit – II & Unit –III** each.*

	Unit – I	Marks	BT*	CO*	PO*
1. a)	What are the major public acceptance challenges associated with biotechnology, especially in areas like genetically modified organisms (GMOs)?	10	L*1	1	1.1.2 1.2.1
b)	Analyze how legal frameworks in different countries regulate the use of biotechnology in sensitive areas such as genetic modification.	10	L4	1	1.1.2
2. a)	What are the ethical implications of using cloning technologies in agriculture and medicine?	10	L1	1	6.1.1
b)	Analyze the role of bioethics in preventing the misuse of biotechnology for harmful purposes such as biological warfare.	10	L4	1	6.1.1
	Unit – II				
3. a)	List the biosafety levels (BSL) and match each level with its corresponding risk group (RG).	10	L1	2	6.1.1
b)	Name four components of risk assessment and outline their purpose in risk management.	10	L1	2	6.1.1
4. a)	Provide an explanation that demonstrates understanding of hazard assessment and exposure management.	10	L2	2	1.1.2
b)	Identify and recall the main principles behind GMP in a biological laboratory.	10	L3	3	1.1.2
5. a)	Illustrate the significance of the Cartagena Protocol on Biosafety in regulating the movement of genetically modified organisms (GMOs).	10	L2	3	1.1.2
b)	Provide a practical example of conducting a risk assessment, highlighting key considerations for rDNA technology.	10	L2	3	6.1.1
	Unit – III				
6. a)	Outline the key steps involved in filing a patent, differentiate between provisional and non-provisional patent applications	10	L2	4	1.2.1
b)	What is the TRIPS Agreement, and how does it standardize the protection of intellectual property across WTO member countries?	10	L1	4	1.2.1
7. a)	List the main activities and objectives of the World Intellectual Property Organization (WIPO).	10	L1	4	6.1.1
b)	What precautions should be taken when patenting an invention, particularly regarding disclosure and non-disclosure of information?	10	L1	5	6.2.1
8. a)	Illustrate the process of patent filing and the importance of patent searches and databases in the procedure.	10	L2	5	1.2.1
b)	Classify various IPRs and infer why Plant Breeders' Rights might be the best option for agricultural innovations.	10	L2	5	1.2.1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
